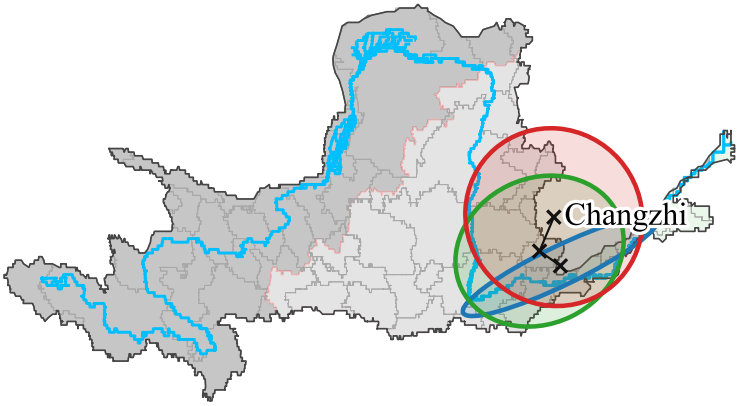
(a) Water - Grid



(b) Water - City



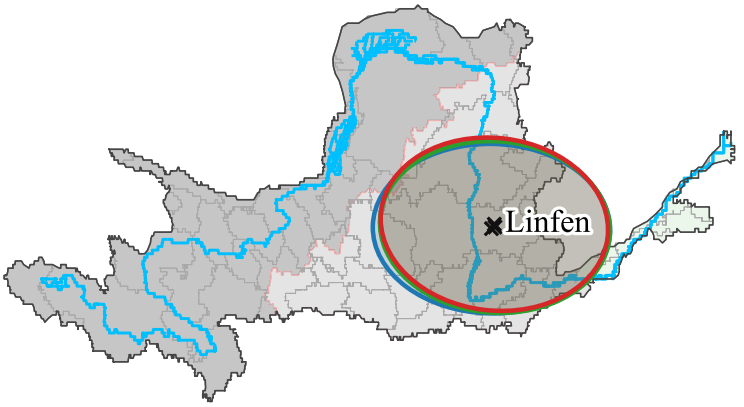
(c) Energy - Grid



(d) Energy - City



(e) Food - Grid



(f) Food - City



2010 2015 2020 Ellipse centre The Yellow River

Upper reaches Middle reaches Lower reaches

(g) SDE centre displacement from 2010 to 2020

Scale	System	2010 Centre	2020 Centre	Movement direction	Distance, km
Land-use-based	Water	Qingyang	Qingyang	North-east (47.0°)	19.733
Land-use-based	Energy	Jiaozuo	Changzhi	North (357.0°)	133.987
Land-use-based	Food	Linfen	Linfen	North-east (37.7°)	14.599
City-scale	Water	Yan'an	Yan'an	East (67.6°)	14.511
City-scale	Energy	Yan'an	Yan'an	South-west (237.5°)	12.671
City-scale	Food	Yan'an	Yan'an	East (102.5°)	20.437